

Suryaansh Jain

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EDUCATION

UNIVERSITY OF MASSACHUSETTS AMHERST | *M.S. in Computer Science* | GPA: 3.95/4.0 2025 – 2027

- Coursework: Adv. Topics in ML, Reinforcement Learning, Probabilistic Graphical Models, Robotics, Industry Research (Adobe), Complexity Theory, Systems for Machine Learning*, Information Retrieval*, Introduction to Simulation*

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *B.Tech in Computer Science* | GPA: 9.41/10 2021 – 2025

- Selected coursework: Foundations of ML, Deep Learning, Adv. Topics in DL, RL, Computer Vision, Convex Optimization, Information Theory, Spectral Graph Theory, Game Theory, Theory of Computation, Computational Complexity, Approximate Algorithms, Cryptology, Quantum Computing

PREPRINTS

Zero-shot Fine-grained Image Captioning with VLAs 2026

S. Jain, Vishal G, David S. Yoon. Manuscript being submitted to AAAI 2027; first author. [†]

Fantastic ϕ 's and where to find them: Reward Based Abstractions for STAR 2026

S. Jain, S. Chaudhari, P. S. Thomas. Manuscript being submitted to AAAI 2027; first author. [†]

Beyond Consensus: Mitigating the Agreeableness Bias in LLM Judge Evaluations Oct. 2025

S. Jain, U. Z. Ahmed, S. Sahai, B. Leong. [arXiv:2510.11822]

A Bound for the Cops and Robber Problem via k -Component Order Connectivity Dec. 2024

S. Jain et al. [arXiv:2412.02511]

RESEARCH EXPERIENCE

UMASS HUMAN-CENTERED ROBOTICS LAB | *Graduate Researcher* Amherst, MA

Zero-Shot Generalizable Multi-Task World Models Feb. 2026 – present

- Built single world model jointly trained on **71 visual control tasks** (OGBench, Meta-World, RoboDesk; action spaces 2–105 dim) over a frozen **DINOv2** backbone with a universal action encoder (zero-pad + binary mask).
- Added dual natural-language task/robot conditioning via frozen **MiniLM** sentence embeddings, enabling zero-shot transfer to new tasks/embodyments without retraining; matched LeWM single-task specialist on cube within 3 SR points at $\sim 18\times$ less compute.

UMASS HUMAN-CENTERED ROBOTICS LAB | *Graduate Researcher* Amherst, MA

VLA Models for Multi-Robot Formation Control Jan. 2026 – present

- Multi-robot coordination using **VLA models** for language- and vision-conditioned formation control; trained in **NVIDIA Isaac Lab**.

ADOBE RESEARCH | *Student Researcher* San Jose, CA (Remote)

Fine-Grained Image Captioning via RL with Detailed Rewards Jan. 2026 – present

- Designed object-aware perception loss extending **PAPo** via **YOLOv13** per-object **KL terms**.
- Designed **SimLoss**, an information theory inspired loss function to enable **self-supervised fine tuning of VLMs**.
- Matched the performance of multi pass methods with $10\times$ improvement in cost and latency.

UNIVERSITY OF MASSACHUSETTS AMHERST | *Graduate Researcher* Amherst, MA

Fantastic ϕ 's and where to find them: Reward Based Abstractions for STAR Sep. 2025 – present

- Designed ϕ -function state abstractions for low variance and bias off-policy estimation. **Beats IS and Model based methods** in all RL environments.

NATIONAL UNIVERSITY OF SINGAPORE | *Visiting Researcher* Singapore

Beyond Consensus: LLM-Judge Bias May 2024 – Oct. 2025

- Quantified **agreeableness bias** across 14 SOTA LLMs ($\sim 96\%$ TPR / $<25\%$ TNR on 366 Python programs); proposed minority-veto + regression calibration, reducing max error to **1.2%** ($2\times$ over best ensemble).

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *Undergraduate Researcher* Hyderabad, India

Cop Number via k -Component Order Connectivity 2023 – 2024

- Strengthened state-of-the-art upper bound on cop number from $|U|/2 + 1$ to $|U|/3 + 9$; first-author manuscript.

- Worked on OTEx: Ownership Transfer and Execution Protocol for Blockchain Interoperability; **Toyota-funded** cross-chain research.

INDUSTRY EXPERIENCE

APPLIED MATERIALS | Machine Learning Intern
SPG Team

Santa Clara, CA
Summer 2026 | 12 Weeks (Ongoing)

- Building **time series forecasting models** to predict the inventory using **Temporal Fusion Transformers** and **N-HiTS** models.

CROW CANYON SOFTWARE | Machine Learning Intern
Nitro Studio AI Team

Remote
Winter 2025 | 6 weeks

- Engineered **Agentic RAG** systems with **LangGraph**, leveraging custom agent workflows and PDF-based knowledge retrieval to build customer support chatbots for Nitro Help Desk.

BRYT SCHOOLS | ML & Software Engineering Intern
Tutor Development Team

Remote
Summer 2025 | 10 weeks

- Shipped LLM-powered conversational chatbot: fine-tuning, eval, production deployment across **100+ files**.

ACADEMIC PROJECTS

Robustness of Object Detectors to Adversarial Attacks Computer Vision course

Spring 2025

- Reproduced patch- and gradient-based attacks; designed novel variant **Edge-DPAttacks** on **YOLO** and **Faster R-CNN**.

Diffusion + GNNs for Graph Coloring Adv. Topics in DL course

Fall 2024

- Diffusion-based generative framework over graphs; **10% error rate** on 4-colorable graphs, extended to arbitrary k -colorable.

Recommender Systems Using Deep Generative Models Deep Learning course

Spring 2024

- Surveyed SOTA recommender system models like DiffRec, Diff4Rec, and Contrastive Variational AutoEncoder for Sequential Recommendation. Presented these papers and proposed potential improvements.

Drone Motor Failure Detection Inter-IIT Tech Meet

2024

- LSTM detector for motor failure on drone-motor telemetry; deployed in C++ within **ROS2** for real-time inference on **PX4** hardware.

HONORS & AWARDS

- **OnePlus-Oppo Genius+ Scholarship** – 6 students nationwide 2022
- **All India Rank 311** in IIT-JEE Advanced (155K candidates) | **AIR 458** in IIT-JEE Mains (1M candidates) 2021
- Selected for the Indian Chemistry Olympiad **Orientation Camp (AIR 27, HBCSE)** 2021
- **KVPY Fellowship**, Government of India – awarded **twice** 2019, 2020

TEACHING & ACTIVITIES

- **Teaching Assistant**, Reinforcement Learning, UMass Amherst 2026
- **Teaching Assistant**, Information Theory and Theory of Computation, IIT Hyderabad 2024, 2025
- **3rd Place (2024) and 2nd Place (2023), Math Bowl (CMI)** – Inter-IIT Tech Meet 2023, 2024
- **Head**, Web Dev Team, *Milan* – IIT Hyderabad’s annual cultural festival 2024

TECHNICAL SKILLS

Languages Python, C/C++, Java, JavaScript, Bash, SQL, Rust, Dart, LaTeX

ML & Systems PyTorch, TensorFlow, JAX, OpenCV, CUDA, Docker, ROS2, Gazebo, NVIDIA Isaac Lab, MuJoCo

Libraries NumPy, Pandas, scikit-learn, SciPy, Diffusers, CLIP, LangChain, LangGraph, Transformers, Trl, Unsloth, OpenRouter, FastAPI, Django, React, Flask, Flutter, SwiftUI, AWS

[†] Manuscript available upon request. ^{*} Courses in Fall 2026.